

Impact of drug shortages on U.S. health systems

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The American Society of Health-System Pharmacists (ASHP) defines a drug shortage as “a supply issue that affects how the pharmacy prepares or dispenses a drug product or influences patient care when prescribers must use an alternative agent.”¹ Drug shortages can be caused by natural disasters, a lack of available raw or bulk materials, manufacturing difficulties, regulatory issues, recalls of the affected or related products, or changes in product formulations.^{1,2} ASHP and the Food and Drug Administration (FDA) have noted an increase in drug shortages in the past decade.^{3,4} The drug information service at University of Utah Health Care (UUHC), the organization that supports the ASHP drug shortage program, tracked a total of 224 drug shortages between January 1996 and June 2002.^{3,5} In 2007, UUHC identified 129 new drug shortages, with 166 additional new shortages identified in 2009.⁶ In 2010, the number of drug shortages

Purpose. A study was performed to quantify the personnel resources required to manage drug shortages, define the impact of drug shortages on health systems nationwide, and assess the adequacy of information resources available to manage drug shortages.

Methods. An online survey was sent to the 1322 members of the American Society of Health-System Pharmacists who were identified as directors of pharmacy. Survey recipients were asked to identify which of the 30 most recent drug shortages listed affected their health system, to identify actions taken to manage the shortage, and to rate the impact of each shortage. Employees responsible for completing predefined tasks were identified, and the average time spent by each type of employee completing these tasks was estimated. Labor costs associated with managing shortages were calculated.

Results. A total of 353 respondents completed the survey, yielding a response rate of 27%. Pharmacists and pharmacy technicians spent more time managing drug shortages

than did physicians and nurses. There was a significant association between the time spent managing shortages and the size of the institution, the number of shortages managed, and the institution's level of automation. Overall, 70% of the respondents felt that the information resources available to manage drug shortages were not good. The labor costs associated with managing shortages in the United States is an estimated \$216 million annually.

Conclusion. A survey of directors of pharmacy revealed that labor costs and the time required to manage drug shortages are significant and that current information available to manage drug shortages is considered suboptimal.

Index terms: American Society of Health-System Pharmacists; Communication; Data collection; Drug distribution; Economics; Hospitals; Industry, pharmaceutical; Personnel, pharmacy; Pharmacists, hospital; Pharmacy, institutional, hospital; United States
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reached 211, the highest number recorded to date in a single year.

While many shortages can be effectively managed, considerable per-

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sonnel resources are often required to identify available alternative products or therapeutically equivalent agents and to ensure operational changes are made to accommodate the product change.^{3,7} Pharmacists often spend a large amount of time communicating with manufacturers and wholesalers, providing education to facility personnel on alternative agents, developing or modifying policies or clinical guidelines, and updating electronic databases and medication administration systems.^{1-3,8,9} Although the pharmacy department often plays a leadership role in managing drug shortages, a collaborative effort of all health care providers affected by the shortage is essential.⁷ This is particularly true when the supply of medication cannot meet the clinical need, in which case existing supplies must be allocated to those patients with the greatest need, with alternative therapies identified for other uses. This allocation process requires interdisciplinary collaboration among pharmacy, medical, and nursing staff.³ The amount of time required by pharmacy and nonpharmacy personnel to effectively manage drug shortages can be extensive.^{3,9}

The purpose of this study was to quantify the personnel resources required to manage drug shortages in health systems in the United States. Secondary objectives included defining the extent to which 30 recent drug shortages affected health systems across the country and determining the adequacy of existing resources available for information regarding drug shortages.

Methods

Clinicians at the University of Michigan Health System (UMHS) collaborated with colleagues at ASHP to develop and distribute an online survey using survey software (Qualtrics, Provo, UT). The survey was pilot tested by eight individuals (four health care clinicians at UMHS and four pharmacists at ASHP) using in-

depth interview and cognitive feedback techniques.¹⁰ Revisions were made to improve question clarity and consistency of response. After validity testing, an e-mail invitation explaining the study was sent to ASHP members identified in the ASHP membership database as directors of pharmacy in the United States and its territories, asking for their participation. A link to the online survey instrument was included in the e-mail invitation. An electronic document was provided to participants to facilitate data collection, as the survey required input from various health care professionals within each institution, including pharmacy (e.g., purchasing, operations, drug information), medical, and nursing staff. The survey launched on October 28, 2010, and closed November 16, 2010, with two reminder e-mails sent to encourage participation. This study received an exemption from the University of Michigan's institutional review board for the protection of human subjects.

The Web-based survey collected data in four domains: demographics, impact of recent drug shortages, resource utilization to manage shortages, and the adequacy of currently available information resources on drug shortages. Demographic data included the facility type, size, geographic location, and level of automation of the medication-use process (defined as the number of automated systems that an institution had from a prepopulated list of options). Five questions were included to assess respondents' global perceptions regarding the burden and financial impact of drug shortages, whether shortages led to frustration directed at pharmacy staff, and whether shortages compromised care or changed practice.

Participants were provided with a list of 30 recent drug shortages that had been posted on the ASHP website and were asked to note if the shortage affected their institution and, if so, to

identify actions taken to manage the shortage. In an effort to ensure consistency in responses, four categories of actions were defined: (1) focused inventory tracking (i.e., monitoring and moving stock to avoid a shortage), (2) changes in dispensing practice (i.e., dispensing the same drug entity in a different strength or unit of package), (3) allocation of remaining supplies (i.e., prioritizing who will receive remaining supplies of the drug), and (4) identification and implementation of alternative therapies (i.e., a different therapeutic agent needed to be substituted). Respondents were also asked to assess the impact that each shortage had on their institution using one of three ratings: low, medium, or high. Specific criteria were provided to assess the impact level based on effort and time spent managing each shortage. "Management" was defined as the collective actual time spent by all involved parties to identify and purchase alternatives, modify databases, stock products, and notify staff. Low-impact shortages were those that were managed in 8 hours or less, medium-impact shortages took 8–16 hours to manage, and high-impact shortages took more than 16 hours. Shortages for which stock was monitored and redistributed in an effort to avert a shortage were also assessed, as these situations can require significant labor resources.

The personnel resources required to manage drug shortages were quantified by job class based on estimates of total time spent on predefined tasks by responsible parties (pharmacists, pharmacy technicians, physicians, and nurses). Tasks were consolidated into five main categories: gather details about the shortage, identify alternatives, manage inventory, communicate information to health care providers, and manage database information systems. Survey participants were asked to indicate the type of employee responsible for each task, including multidisciplinary involvement (i.e.,

physicians, nurses) when appropriate, and the average time each discipline spent (in hours per week) managing drug shortages in calendar year 2010. The time spent by others within the health system who may be involved in the process but are not included in one of the four predefined job groups (e.g., informatics specialists who are not pharmacists, pharmacy technicians, physicians, nurses) was not assessed.

In an effort to assess the adequacy of information sources available regarding drug shortages and to determine where opportunities for improved communication with health systems exist, the survey included specific questions about the various tools currently available to clinicians (i.e., ASHP drug shortage website, FDA drug shortage website, wholesaler and group purchasing organizations, and direct communication with manufacturers). Participants were asked to assess the usefulness of each information source using a 5-point scale, where 1 = very poor and 5 = very good. Rating characteristics included timeliness of shortage information, completeness of information (reason and anticipated duration), and whether alternatives were suggested. Participants were also asked to rate the overall adequacy of the information available to help manage drug shortages.

Data were analyzed with SPSS Statistics, version 19.0 (SPSS Inc., Chicago, IL). Descriptive statistics and frequency tables were used to characterize hospital characteristics, the personnel resources required to manage drug shortages, and the usefulness of existing information sources. Due to the presence of a few extreme outliers, median and interquartile ranges were used to report the number of hours spent per week managing drug shortages. A sensitivity analysis in which the data were analyzed with and without the outliers revealed that the outliers had no effect on the median values.

As such, all data points were included in the data analysis. Mann–Whitney and Kruskal–Wallis nonparametric analyses were used to assess whether a significant relationship existed between the median total time spent per week and each of the following: size of the organization, number of shortages experienced, and level of automation present in the organization. The a priori level of significance was 0.05. Frequency tables were used to outline the breadth of the 30 recent drug shortages. The association between the number of shortages experienced by a health system and its size or geographic location was evaluated using chi-square analysis.

Total labor costs required to manage shortages were calculated by applying national average salary

information^{11,12} to the median hours spent by job class, which was obtained from the survey. An estimate of the national annualized labor costs associated with managing drug shortages was calculated by applying the study data to the 2011 actual national number of health systems, by hospital size, available from the SDI Health Hospital Database (SDI Health, Yardley, PA).

Results

The survey was sent to 1322 ASHP members who identified themselves as directors of pharmacy in the ASHP membership database. All respondents who answered at least one question on the survey ($n = 353$, 27% response rate) were included in the study (Table 1); all 50 states were rep-

Table 1.
Demographics of Survey Respondents^a

Characteristic	No. (%) Respondents
Type of hospital ($n = 300$)	
Community (not-for-profit)	192 (64)
For-profit	38 (13)
University	19 (6)
Government	9 (3)
Other	42 (14)
No. staffed beds ($n = 299$)	
<100	87 (29)
100–199	54 (18)
200–399	79 (27)
≥400	64 (21)
Outpatient beds only	15 (5)
No. pharmacist FTEs ($n = 286$)	
≤10	150 (53)
11–25	78 (27)
26–50	29 (10)
>50	29 (10)
No. pharmacy technician FTEs ($n = 285$)	
≤10	138 (49)
11–25	95 (33)
26–50	26 (9)
>50	26 (9)
Geographic region ($n = 353$)	
West	73 (27)
Midwest	94 (19)
South	118 (33)
North	67 (21)
Outside United States (Virgin Islands)	1 (<1)

^aFTEs = full-time equivalents.

resented. The distribution of respondents by facility size differed from the national distribution of acute care facilities, with the survey population having a higher representation of larger hospitals (400 or more beds) compared with national data (21% versus 7%, respectively).

Time spent managing drug shortages. The time spent by pharmacists, pharmacy technicians, physicians, and nurses managing drug shortages during calendar year 2010 is shown in Table 2. The collective time for each of the five predefined activities (gather details, identify alternatives, manage inventory, communicate information, and manage information or data systems) is reported. The pharmacy department (pharmacists and pharmacy technicians) spent more time managing drug shortages (nine and eight hours a week, respectively) than did physicians and nurses, who spent less than one hour per week performing this task. Physicians

and nurses were reported to spend most of their time identifying drug alternatives and communicating information about the shortage to their colleagues. There was a significant association between the amount of time spent managing shortages and the hospital size (i.e., the amount of time increased as the institution size increased). A similar significant association existed between the time spent managing drug shortages and the total number of shortages or the level of automation (Table 2).

The estimated annual labor cost of managing drug shortages based on institution size is shown in Table 3. The median survey data by hospital size applied to actual national data on health systems (by hospital size) resulted in an estimated annual labor cost of \$216 million (interquartile range, \$114.7 million–\$483.1 million) for all health systems nationwide.

Few institutions ($n = 14$, 5%) reported adding incremental staff

in the previous two years to manage drug shortages. The positions most often added included pharmacists (median, 0.1 full-time equivalent [FTE]; interquartile range, 0–0.75 FTE) and pharmacy technicians (median, 0.5 FTE; interquartile range, 0–0.8 FTE). Reallocation of existing staff to allow time for individuals to focus on managing shortages was reported by 32% of respondents.

Impact of drug shortages. The number of drug shortages experienced in calendar year 2010 is shown in Figure 1. The association between increasing hospital size and number of shortages experienced was significant ($p = 0.001$). Nearly half (47%) of institutions with 400 or more beds experienced more than 30 drug shortages during calendar year 2010. Responses to five global assessment questions are shown in Figure 2; at least 50% of respondents indicated agreement with each of these questions. Over 90% of respondents

Table 2.
Time Spent Managing Drug Shortages

Hospital Characteristic	Median No. Hours per Week (Interquartile Range)			
	Pharmacists	Pharmacy Technicians	Physicians	Nurses
Overall	9 (5–20)	8 (3–17)	0.5 (0–2)	0 (0–2)
Size (no. beds)				
<100 ($n = 87$)	8 (5–12.5)	5 (2–11)	0 (0–1)	0 (0–1)
100–199 ($n = 54$)	9 (5–19.75)	9 (3–17.5)	0.5 (0–2.25)	0 (0–2)
200–399 ($n = 79$)	9 (5–27.5)	10 (5–20)	1 (0–2)	1 (0–2)
≥400 ($n = 64$)	12 (8–28.5)	13 (6–21)	1 (0–4)	1 (0–4.5)
p^a	0.028	<0.0001	<0.0001	0.003
No. shortages experienced				
<10 ($n = 36$)	4.5 (2.25–10)	2.75 (0.6875–6)	0 (0–0.5)	0 (0–0.875)
10–20 ($n = 116$)	8 (4.25–20)	6 (2.25–12)	0 (0–1)	0 (0–1.5)
21–30 ($n = 84$)	11 (5.75–26)	10 (5–21)	1 (0–3.5)	1 (0–2.75)
≥30 ($n = 95$)	10.5 (6.12–23.75)	13 (6–21.5)	1 (0–2.75)	0.125 (0–2)
p^a	<0.0001	<0.0001	<0.0001	0.034
Level of automation (no. automated systems present)				
0 ($n = 53$)	4.5 (1.13–22.5)	1.63 (0–22.25)	0 (0–1.25)	0 (0–2.5)
1–4 ($n = 74$)	7 (4.5–17)	5 (2–10)	0 (0–1)	0 (0–1)
5–8 ($n = 115$)	10 (5–19.25)	8.5 (3.25–15.75)	0.5 (0–2)	0 (0–2)
9–12 ($n = 92$)	10 (5–26.5)	12 (5–21)	1 (0–2.5)	1 (0–3.5)
13–16 ($n = 19$)	14.29 (9.5–28.5)	14 (6.5–18.25)	1 (0–5)	0 (0–3.25)
p^a	0.024	<0.001	0.002	0.022

^aKruskal–Wallis analysis for association between time spent managing shortages and increasing bed capacity, number of shortages, and level of automation.

agreed that currently an increased work burden and increased costs were associated with drug shortages compared with those experienced two years before the survey.

The percentage of institutions affected by each of the 30 recent drug shortages is shown in Figure 3. At least 50% of respondents indicated that their organization was affected by 7 of the drugs; 3 drugs affected over 80% of the organizations (dextrose 50% syringes, epinephrine injection, and succinylcholine injection). Only 4 (1%) respondents

reported that their organization was not affected by any of the 30 drug shortages. Of the 30 drugs, the shortage of 17 was deemed medium or high impact by at least 60% of respondents. In contrast, the shortage of only 2 drugs was deemed low impact by at least 60% of respondents. There was no association between geographic location and number of shortages experienced, and drug shortages did not vary by geographic location. Each of the predefined strategies for managing drug shortages was used for all

drugs, indicating a multiprocedural approach to management. Respondents indicated that they were using an average 1.9 (range, 1.6–2.2) of the 4 predefined strategies per shortage experienced. Overall, respondents indicated using focused inventory tracking most frequently, with its use averaging 65% across all 30 drug shortages. Other strategies were used less often: changing dispensing practices and allocating remaining supplies, 44% each; and identifying and implementing alternative therapies, 39%.

Table 3.

Estimated Annual Labor Costs Associated With Managing Drug Shortages

Hospital Size (No. Beds)	Median Annual Labor Cost ^a (Interquartile Range) (\$)				
	Pharmacists	Pharmacy Technicians	Physicians	Nurses	Total
<100	21,753 (13,595–33,989)	3,520 (1,408–7,745)	0 (0–4,741)	0 (0–1,697)	25,273 (15,004–48,172)
100–199	24,472 (13,595–53,702)	6,337 (2,112–12,321)	2,371 (0–10,668)	0 (0–3,394)	33,179 (15,708–80,085)
200–399	24,472 (13,595–74,775)	7,041 (3,520–14,082)	4,741 (0–9,483)	1,697 (0–3,394)	37,951 (17,116–101,733)
≥400	32,629 (21,753–77,494)	9,153 (4,225–14,786)	4,741 (0–18,965)	1,697 (0–7,635)	48,220 (25,977–118,880)

^aCost calculated by annualizing the weekly median time spent managing drug shortages and applying national average salary data (\$52/hr for pharmacists, \$14/hr for pharmacy technicians, \$91/hr for physicians, and \$33/hr for nurses).^{11,12}

Figure 1. Number of shortages affecting health systems by hospital size in calendar year 2010.

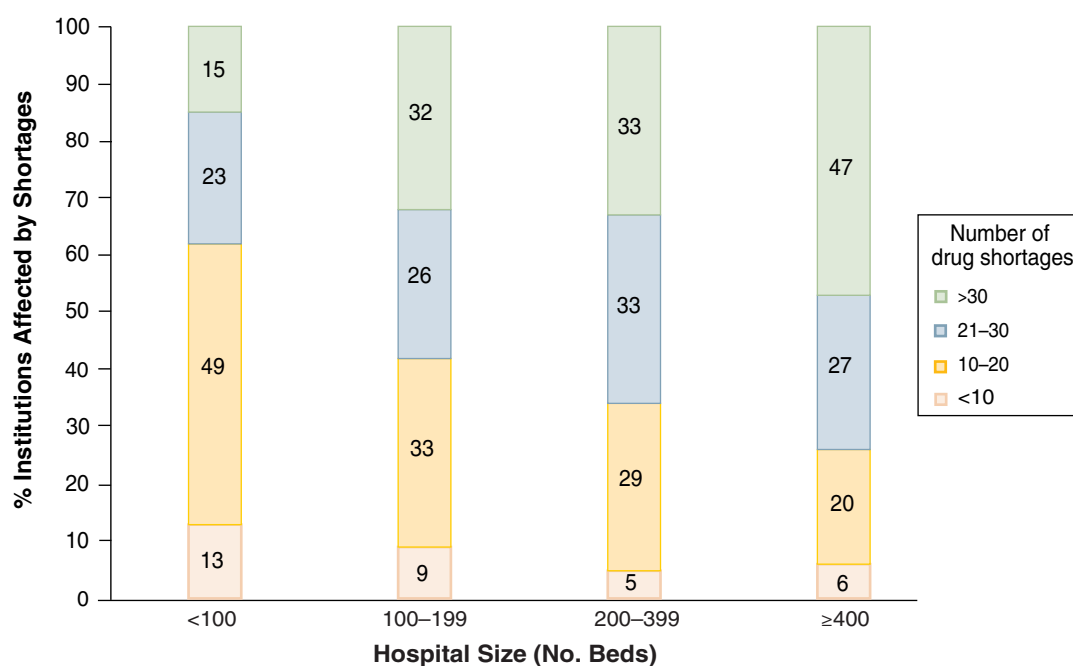
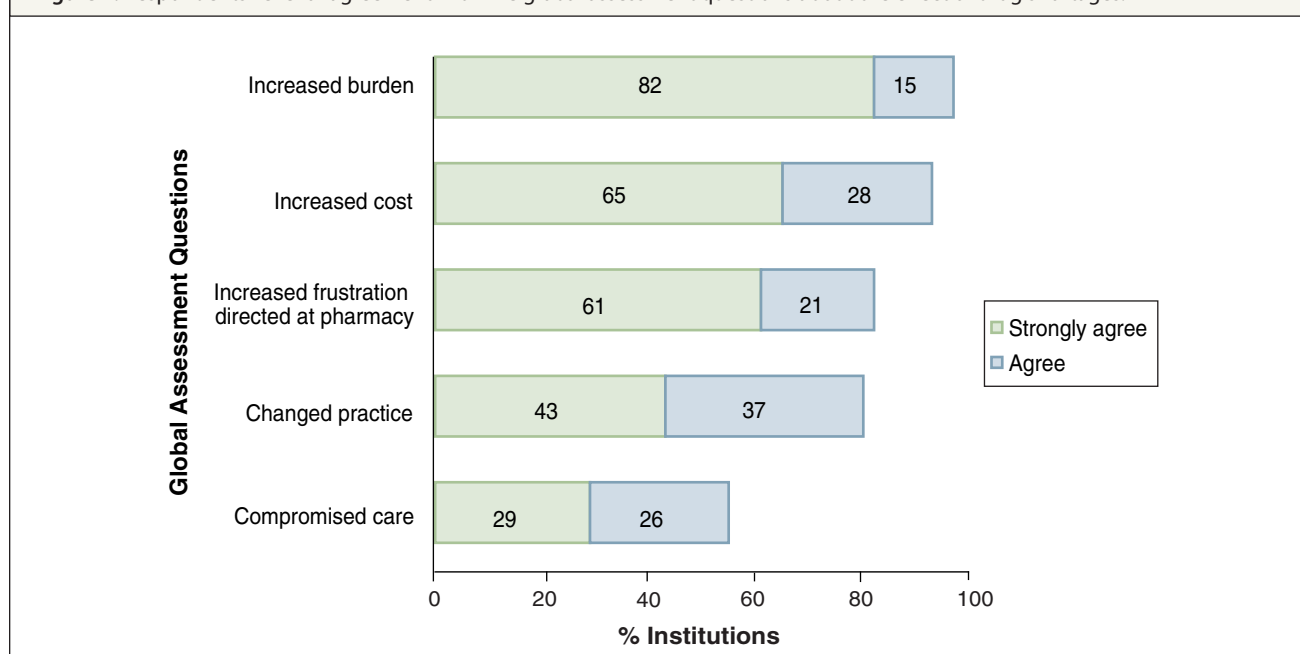


Figure 2. Respondents' level of agreement with five global assessment questions about the effect of drug shortages.



Communication about drug shortages was distributed to health care providers through multiple means, with e-mail (88%), clinician-to-clinician discussions (75%), and newsletters (57%) being the most common. While daily or weekly communication to pharmacy staff was common (61%), most respondents indicated no set schedule for communication to nonpharmacy staff (54%).

Drug shortage information resources. Most respondents (76%, $n = 231$) reported having established policies, guidelines, or processes for managing drug shortages; 59% of them reported using the ASHP Guidelines on Managing Drug Product Shortages in Hospitals and Health Systems¹ to formulate their drug shortage policy. The ASHP guidelines were determined to be helpful in adequately managing drug shortages by 56% of respondents; 22% were not aware of the guidelines. Most respondents reported using multiple resources to gather information about existing drug shortages, including the ASHP drug shortages website (89%), the

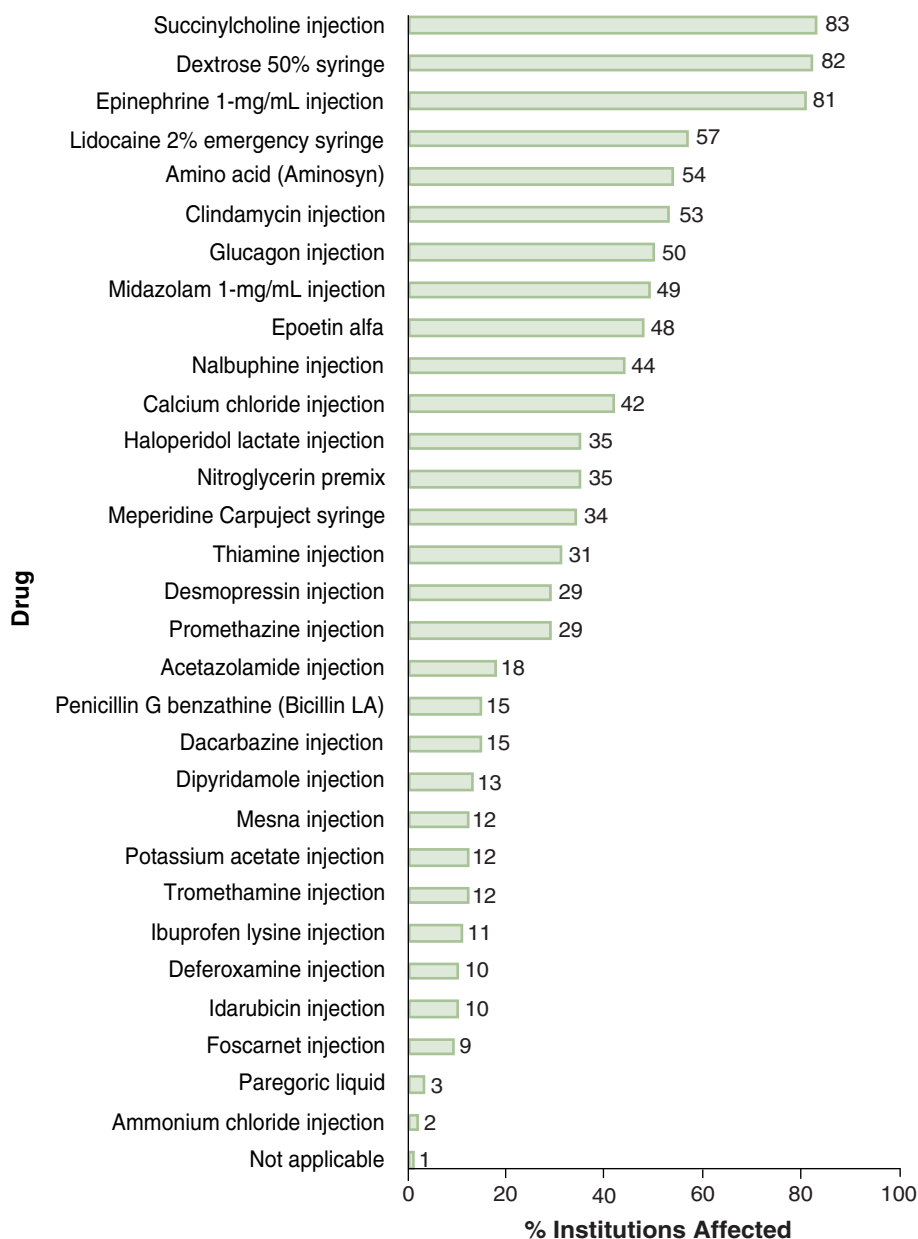
wholesaler website (84%), the group purchasing organization website (74%), the FDA drug shortages website (72%), or direct communication with the manufacturer (67%). Table 4 shows respondents' ratings of the usefulness of each of these information sources based on the mean score (using a 5-point Likert scale) assessing three variables: timeliness of the information, the reason for and anticipated duration of the shortage, and suggestions for alternatives. The ASHP drug shortages website was consistently rated the most useful in each of the three domains. Overall, 70% of respondents felt the information available to manage drug shortages was less than good (rated 1–3 on a 5-point Likert scale). Only 2% of respondents found the current state of information to be very good (rated 5).

Discussion

Drug shortages present an ongoing challenge for health care providers. When shortages occur, health systems must act quickly to identify and obtain alternative products

to prevent disruptions in patient care. The clinical and economic impact of drug shortages was first reported in 2004 in a national survey conducted by ASHP and The Johns Hopkins Hospital.⁹ At that time, pharmacists were spending a median of 3 hours per week managing shortages, with an additional 2.5 hours per week spent by nonpharmacist personnel. In our study, pharmacists and pharmacy technicians managed drug shortages for a median of 9 and 8 hours per week, respectively, suggesting that the time being spent by pharmacy and nonpharmacy personnel has increased since 2004. Likely explanations for this increase include a greater number of drug shortages, an increase in the number of drug shortages that require identification of therapeutic alternatives, and an increase in the complexity of health care delivery systems. Currently, many institutions have multiple automation systems available, including computerized physician order entry, bedside bar-code-assisted medication administration, computerized perpetual inventory systems,

Figure 3. Percentage of institutions affected by each of the 30 most recent drug shortages listed on the ASHP website in calendar year 2010.



automated dispensing cabinets, and cart-fill robots. While these automated systems allow for improvements in the drug-ordering and delivery system and provide tools to manage drug shortages that were not easily available with manual systems (e.g., alerting prescribers to alternative agents at the time of order entry),

they also present new challenges. The databases associated with these systems require human resources to manage change and ensure connections among automation systems that do not communicate with each other. In an effort to ensure consistent management of drug shortages, some institutions have created

positions dedicated to addressing shortages, while others manage them on an ad hoc basis using personnel resources available at the time.³ Drug shortage management activities often must take precedence over other responsibilities, shifting the work of clinicians away from high-impact, high-value activities (e.g., direct pa-

Table 4.
Respondents' Ratings of Existing Information Sources on Drug Shortages

Rated Characteristic	Mean Rating ^a			
	ASHP Website	FDA Website	Wholesaler Website or Communication	GPO Website or Communication
Timeliness	3.93	3.40	3.38	3.42
Reason for and duration of shortage provided	3.80	3.23	2.66	3.23
Suggested alternatives provided	3.97	2.74	1.81	3.04
				3.02
				2.78
				1.92

^aBased on a 5-point Likert scale, where 1 = very poor, 2 = poor, 3 = neutral, 4 = good, and 5 = very good. ASHP = American Society of Health-System Pharmacists, FDA = Food and Drug Administration, GPO = group purchasing organization.

tient care, therapeutic drug monitoring, optimal delivery of medication therapy) to that of ensuring availability of a supply of a critical medication or its alternative.

While the labor costs associated with managing shortages were not reported in the 2004 ASHP national survey, the annual increase in drug acquisition costs required to purchase alternative products was estimated at \$1 million to the U.S. health care system.⁹ In a more recent analysis, Premier Healthcare Alliance estimated that the need to purchase more expensive generic drugs or therapeutic alternatives caused by drug shortages could be costing U.S. hospitals at least \$200 million annually.¹³ When \$216 million—for annual labor costs associated with managing drug shortages, as estimated in the current study—is added, the economic burden created by drug shortages becomes even more apparent.

Economic effects are not the only concerns associated with the increase in drug shortages. There are also considerable medication safety concerns. In July 2010, the Institute for Safe Medication Practices surveyed 1800 health care professionals to better understand the frustrations and patient safety concerns associated with drug shortages.¹⁴ Clinicians reported increased risks of adverse effects during a drug shortage, in part because physicians may be prescribing an alternative agent with which they are unfamiliar. There can also be compromised clinical outcomes if a less-efficacious agent must be used or if no good therapeutic alternative is available. The potential for medication errors also increases; 35% of respondents reported that their facility experienced a near-miss error due to a drug shortage in the previous year.^{6,14} Similar concerns have been expressed in other recent reports.^{13,15} Our results confirm what prior surveys have shown—pharmacy directors believe that the burden of drug shortages continues

to increase and that drug shortages have changed clinical practice and compromised patient care.^{9,13} There is also concern that drug shortages are compromising interdisciplinary relationships as frustration directed toward pharmacy personnel by other health care providers is increasing.

In order to successfully and efficiently manage drug shortages, it is critical that pharmacists have access to reliable and timely information. In 2001, ASHP began to provide guidelines for managing drug shortages, outlining suggested processes that can reduce the challenges posed.¹ ASHP also partnered with UUHC to develop the drug shortage website. Later, additional resources announcing information about drug shortages became available, including the FDA drug shortage website. This site, though, is limited in that it focuses only on medically necessary drugs, defined as “any drug product used to treat or prevent a serious disease or medical condition for which there is no other adequately available drug product that is judged by medical staff to be an appropriate substitute.”¹⁶ Despite the availability of these and other resources, health systems are often unaware of drug shortages until they can no longer purchase products, making it difficult for them to develop an effective management strategy.^{7,17} While the actions by ASHP, FDA and others offered support to clinicians, they did not address the growing concern about the increased number of drug shortages and the negative effect these shortages were having on health care in this country.

Because of the growing concern about drug shortages, on November 5, 2010, a drug shortage summit was convened by ASHP in conjunction with the American Society of Anesthesiologists, the American Society of Clinical Oncology, and the Institute for Safe Medication Practices. The goal was to provide strong, effective leadership and a concerted effort

by all involved parties to ensure the supply of medications in our health care systems.⁶ At this meeting, many of the major stakeholders (pharmaceutical manufacturers, wholesalers, group purchasing organizations, health care providers, the federal government, and national organizations) discussed the scope and causes of drug shortages. The summit resulted in 20 recommendations in four key domains to address various factors felt to be contributing to drug shortages: regulatory and legislative, raw material sourcing and manufacturing, business and marketing, and product distribution. Work groups consisting of summit participants and other key stakeholders were established to prioritize and develop detailed action plans for summit recommendations that are most likely to achieve success in addressing root causes of drug shortages.

In February 2011, the Preserving Access to Life-Saving Medications Act was introduced.^{18,19} This act aims to give FDA new tools to deal with potential shortages by requiring manufacturers to report actual or potential interruptions in production that could result in a drug shortage, as well as the development of criteria to identify drugs vulnerable to a shortage to ensure the continuity of the drug supply. Drugs identified as vulnerable would trigger the need for contingency plans for their manufacturing. The bill would require FDA to revise the definition of *medically necessary* to account for drug-use factors. Currently, FDA does not investigate shortages that are temporary and self-limiting or that are limited to only a specific strength or package size of a drug product, despite the fact that such shortages can be very problematic to health systems.² The proposed legislation strives to improve communication among FDA, manufacturers, health systems, and other stakeholders by providing advance notification of potential shortages.

Our study had several limitations. The response rate was relatively low, and the survey was only distributed to those individuals identified in the ASHP membership database as directors of pharmacy. The demographic distribution of respondents based on institution size was heavily weighted to larger community hospitals relative to the national distribution of health systems. The data were corrected to match national hospital distribution when estimating the total labor costs associated with managing drug shortages. In addition, respondents provided estimates of time based on recall, which could have underestimated or overestimated the time each health care professional spent managing drug shortages. Further, it is possible that the input of individuals key to the process of managing shortages was missed (e.g., informatics support personnel managing the institution's automation who may not be pharmacists or pharmacy technicians). Despite these limitations, the results of this study provide valuable information on the effect that drug shortages are having on personnel resource utilization in health systems nationwide.

Conclusion

A survey of directors of pharmacy revealed that labor costs and the time required to manage drug shortages are significant and that current information available to manage drug shortages is considered suboptimal.

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